

# **PERSONAL EXPENSE SPLITTER**

**DBMS Based Application**

On

UCS4001

DATABASE MANAGEMENT SYSTEM



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

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**Submitted To:**

Mr. Aditya Bangiya

**Submitted By:**

Roll No. : 2410030651

Std Name: Srishty Agarwal

Section: 2CSE4

## 1. Project Name

Personal Expense Splitter: - A database-driven program that tracks settlements and computes individual shares automatically to assist organisations in effectively managing shared expenses.

## 2. Problem Statement

Managing shared spending can be challenging in everyday scenarios like group travel, shared housing, or student living. Keeping track of how much each member owes might be challenging when one person covers the cost of food, rent, or travel tickets. Calculations done by hand frequently result in misunderstandings, disagreements, or mistakes. The suggested Personal Expense Splitter system offers a dependable and organised solution by utilising DBMS concepts. Using a relational database, the system allows users to create groups, track shared spending, automatically calculate individual shares, and keep accurate settlement records.

## 3. Objectives

- To design and implement a structured relational database for managing shared expenses.
- To store user and group details securely.
- To record expense transactions with accurate payment information.
- To calculate individual shares automatically.
- To maintain proper records of settlements between users.
- To apply normalization and database constraints to ensure data integrity.

## 4. DBMS Technology Used

|                     |                                            |
|---------------------|--------------------------------------------|
| Database Model      | Relational Model                           |
| Database System     | MySQL / PostgreSQL                         |
| Query Language      | SQL                                        |
| Normalization Level | Up to Third Normal Form (3NF)              |
| Constraints Used    | Primary Key, Foreign Key, NOT NULL, UNIQUE |

RDBMS is selected because it ensures structured data storage, reduces redundancy, and provides reliable transaction handling, which is essential for financial data.

## 5. Dataset Description

This project uses an application-generated dataset. When users register, form groups, add costs, and document settlements, data is generated dynamically.

The dataset includes:

- User information
- Group information
- Expense records
- Split calculations
- Settlement transactions

## 6. Database Tables

Users Table

- UserID (PK)
- Name
- Email
- PasswordHash
- CreatedAt

Groups Table

- GroupID (PK)
- GroupName
- CreatedBy (FK)
- CreatedAt

GroupMembers Table

- GroupID (FK)
- UserID (FK)

- Role

#### Expenses Table

- ExpenseID (PK)
- GroupID (FK)
- PaidBy (FK)
- TotalAmount • Title
- ExpenseDate

#### ExpenseSplits Table

- SplitID (PK)
- ExpenseID (FK)
- UserID (FK)
- OwedAmount
- PaidAmount

#### Settlements Table

- SettlementID (PK)
- GroupID (FK)
- PaidBy (FK)
- PaidTo (FK)
- Amount
- SettlementDate

### 7. Relationships Between Tables

- A user can belong to multiple groups.
- A group can contain multiple users.
- A group can have multiple expenses.
- Each expense can be split among multiple users.

- Users can settle outstanding balances with each other.

## 8. Future Scope

- Development of a mobile application version.
- Integration with digital payment systems such as UPI.
- Addition of graphical dashboards for expense analysis.
- Multi-currency support.
- Cloud-based deployment.
- Enhanced security features like two-factor authentication.

## 9. Conclusion

The Personal Expense Splitter project serves as an example of how DBMS technologies may be used to solve actual financial management issues. Transparency, correctness, and effective expenditure tracking are guaranteed by the system, which arranges users, groups, spending, and settlements into organised relational tables. Additionally, the project offers a scalable framework for upcoming improvements and practical implementation.